

Article

Study of the Ciprofloxacin Adsorption of Activated Carbon Prepared from Mangosteen Peel

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Abstract: In this study, activated carbon derived from mangosteen peel (abbreviated as ACMP) was successfully fabricated. This as-prepared ACMP possessed graphite and had a porous structure with a specific surface area of 419.8554 m²/g. Investigations of the influencing factors on the ciprofloxacin (CIP) adsorption capability of the ACMP based on the static adsorption method showed that: adsorption equilibrium can be achieved in 60 min, the optimal pH for CIP adsorption was a pH of 6, and the optimal ratio between the material mass and solution volume was 3 g/L. The CIP adsorption process of the ACMP follows the Langmuir, Freundlich, Tempkin, Elovich, and Redlich–Peterson isotherm adsorption models. The maximum adsorption capacity calculated according to the Langmuir isothermal adsorption model for the CIP of the ACMP was ($q_{\max} = 29.76$ mg/g). The CIP adsorption process of the ACMP followed the apparent quadratic kinetic equation, as well as spontaneous, endothermic, physical, and chemical adsorption. The adsorption rate was governed by membrane diffusion.

Keywords: activated carbon; ciprofloxacin; adsorption; aqueous environment; mangosteen peel



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1. Introduction

Ciprofloxacin (CIP) is a fluoroquinolone antibiotic which has been widely employed in the treatment of infections in humans and animals in recent decades. Chemically, CIP is a 1-cyclopropyl 6-fluoro-4-oxo-7-piperazine-1-yl-quinolone-3-carboxylic acid, which has an extended aromatic region and functional groups suitable for hydrogen binding. CIP can possibly enter an aqueous environment via either the incomplete metabolism of the body or wastewater from drug manufacturers [1]. The concentrations of CIP can be detected in wastewater and surface water at concentrations of one-hundred ng/L to µg/L [2]. The presence of CIP in wastewater and surface waters, even at trace concentrations, can still have negative effects on environmental ecology and human health [3].

Thus, various chemical reagents and physical and chemical methods have been developed to remove CIP from wastewater, such as adsorption [4–16], photocatalysis [17–21], internal electrolysis [22], ozonation, oxidation [2], flocculation [23], and microbial treatment [24]. Among these methods, adsorption has been proven to be an effective approach because of its ease of operation, low cost, and high efficiency.

Activated carbon is an adsorbent commonly used in practical applications due to its well-developed pore size, the reactivity of its surface functional groups, and its large surface area, which provides more binding sites for adsorption [25]. Currently, the research

direction for manufacturing activated carbon from agricultural by-products is of great interest to many scientists because of the available, abundant, and cost-effective raw materials. Mangosteen, scientifically known as *Garcinia mangostana* L., is a plant belongs to the *Clusiaceae* family. Mangosteen is primarily grown in Southeast Asia, Sri Lanka, and India. In addition, this fruit also appears in some tropical South American countries, for example, Colombia and Puerto Rico. Vietnam is also has probable land for cultivating mangosteen. However, mangosteen only grows well in the South Central, Southeast, and Southwest regions of Asia. Particularly, the Southwest region (the Mekong River Delta) has the largest growing area of mangosteen, approximately 4.9 thousand hectares, with an output of approximately 4.5 thousand tons. Mangosteen peel is the discarded part of mangosteen fruit, containing lignin and hemicellulose compounds, which are primary materials for fabricating activated carbon. Therefore, activated carbon derived from mangosteen peel (ACMP) has been synthesized and applied as a material to treat aqueous environments contaminated with metal ions and dyes by many scientists [26–35]; nonetheless, no studies in the literature have ever reported CIP adsorption via mangosteen peel derived activated carbon.

When making activated carbon from mangosteen peel, the authors used chemical agents such as K_2CO_3 [30], KOH [31], H_3PO_4 and KOH [34], and H_3PO_4 [35] combined with physical agents such as temperature (500–900 °C [30,31,34,35]) and CO_2 [29].

Up to now, no author has studied CIP adsorption on activated carbon materials derived from mangosteen peel. According to previous research papers, the adsorption of CIP on biological activated carbon materials is usually physical adsorption, and the adsorption process is endothermic [7]. No author has studied the Elovich isotherm adsorption model and calculated the activation energy of CIP adsorption on biological activated carbon [7].

Therefore, in this work, we introduce a method to make activated carbon from mangosteen peel through thermal activation at 500 °C for 2 h, followed by chemical activated by $ZnCl_2$. We studied the adsorption isotherm models of Langmuir, Freundlich, Tempkin, Elovich, and Redlich–Peterson, as well as the kinetics, chemical thermodynamics, and activation energy of CIP adsorption processes using the as-fabricated activated carbon.

2. Materials and Methods

2.1. Materials

2.1.1. Fabrication Procedure of Activated Carbon Derived from Mangosteen Peel

Mangosteen peel was collected, washed thoroughly with tap water and distilled water several times to remove all dirt particles, dried, crushed, and placed in a furnace (Carbolite Sheffield, UK, LMF4) at 500 °C for 2 h, with a ramping rate of 10 °C/min. Under such anoxic conditions, the material was thermally decomposed into porous carbon materials and hydrocarbon compounds. The sample was then taken out, cooled at room temperature, and washed with distilled water until its pH was 7. The sample was then dried at 105 °C to constant mass, ground, and sieved to a size of $d = 1$ mm. The resulting product is then denoted as MP.

The as-prepared MP was subsequently mixed with solid $ZnCl_2$ with a mass ratio of MP: $ZnCl_2$ as 2:1 by grounding in an agate mortar, and then it was transferred to a 50 mL porcelain crucible with a lid. A small amount of distilled water was added to make a paste, and then it was stirred for 1 h. Subsequently, the samples were calcined to 500 °C for 180 min. After chemical activation, the products were neutralized with a 0.5 M HCl solution. Then, the samples were washed several times with distilled water until their filtrate reached neutral medium, and then they were dried at 110 °C for 24 h. Finally, the activated carbon sample, now abbreviated as ACMP, was crushed and stored in glass vials for further use.

2.1.2. Investigations on the Physical Properties and Surface Characteristics of Activated Carbon MP and ACMP

The morphologies of the MP and ACMP were examined using scanning electron microscopy (SEM) on a JSM-6510LV unit (Jeol, Tokyo, Japan), and the chemical compositions

of the MP ACMP were determined by energy dispersive X-ray spectroscopy (EDS) on the same JSM-6510LV unit (Jeol, Tokyo, Japan). In addition, the surface functional groups of the ACMP were characterized by Fourier transform infrared spectroscopy (FT-IR) on a Neus 670 (Nicolet, Brighton, MO, USA). The crystalline structure of the ACMP was further verified using X-ray diffraction (XRD) on an Equinox 5000 (Thermo Scientific, Paris, France). The specific surface area of the ACMP was determined by the BET method on a Micromeritics-3030 USA. The SEM, EDS, XRD, and BET were conducted at the Institute of Materials, Vietnam Academy of Science and Technology. The FT-IR infrared spectroscopy was measured at the Department of Chemistry, University of Natural Sciences, Vietnam National University, Hanoi.

2.2. Antibiotic Ciprofloxacin

Ciprofloxacin (CIP) is an antibiotic of the fourth quinolone group belonging to the second generation fluoroquinolone antibiotic system. Its molecular formula is $C_{17}H_{18}FN_3O_3$ and its molar mass is 331.346 g. Figure 1 is the structural formula of CIP.

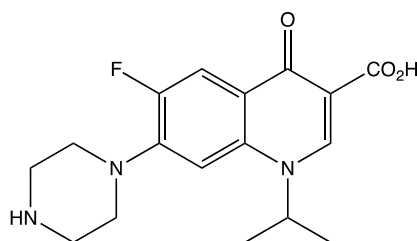


Figure 1. Structural formula of CIP.

2.3. CIP Adsorption Experiments

Experimental factors on CIP elimination, including solution pH, reaction time, mass of the ACMP, initial CIP concentrations, and temperatures, were investigated. To ensure repeatability, each experiment was performed at least 3 times under the same conditions. The results were the average of the 3 experiments. Briefly, a certain amount of ACMP was introduced into each 100 mL Erlenmeyer flask, followed by an additional amount of CIP. A total of 1 mol/L NaOH and 0.1 mol/L HCl were used to adjust the pH of the CIP solution. The flasks were shaken on a shaker with conditions varying according to the experiment (Table 1).

Table 1. Parameters of the experiments.

Experiment/Parameters	pH	Time (min)	Material Mass (g)	CIP Initial Concentration (mg/L)	Temperature (K)
Effect of pH	3 to 10	60	0.075	50	298
Effect of time	6	15–120	0.075	50	298
Effect of mass of initial materials	6	60	0.03–0.2	50	298
Effect of CIP initial concentration	6	60	0.075	50–400	298
Effect of temperature	6	60	0.075	100	303, 313, and 323

The CIP adsorption efficiency of the ACMP was calculated via the following equation:

$$H = \frac{(C_0 - C_e)}{C_0} \times 100\% \quad (1)$$

where H is the adsorption efficiency, C_0 is the initial concentration (mg/L), and C_e is the concentration at adsorption equilibrium (mg/L).

All chemicals ($ZnCl_2$, CIP, NaOH, and HCl) used in the experiment were obtained from Merck and were pure (>99%).

The CIP concentration before and after adsorption was measured on a 02-beam UV-Vis molecular absorption spectrometer, model UH5300, Hitachi, Tokyo, Japan, 2016. The CIP concentration was measured with wavelengths ranging from 190–1100 nm, a scanning speed of 10–6000 nm/s, a wavelength accuracy of ± 0.3 nm, and a noise of < 0.0001 nm.

2.4. Physical Parameters of the Activated Carbon

2.4.1. Moisture Determination

We took 10 g of the ACMP and dried it at 105 °C for approximately 4 h to a constant weight, then weighed and determined its moisture content. The final result was the average of the 3 determinations [26]. The moisture content was calculated using the following equation [27]:

$$\% \text{ Moisture content} = \frac{m_o - m}{m_o} \times 100 \quad (2)$$

where m_o is the initial mass of the AMCP and m is the mass of the AMCP after drying.

2.4.2. Ash Ratio Determination

First, a porcelain cup was heated in a drying oven until its mass was unchanged. Then, it was cooled down and the weight of the cup was measured (m_1). Next, 1.00 g of the AMCP (dried at 80 °C for 24 h) was added into the cup, and we measured the cup containing the AMCP (m_2). The samples were calcined in a furnace at 600 °C for 4 h. Then, the samples were cooled down to room temperature and weighed again (m_3). The ash content of the ACMP was calculated as follows [27]:

$$\% \text{ Ash content} = \frac{m_3 - m_1}{m_2 - m_1} \times 100 \quad (3)$$

2.4.3. Density Determination

We weighed 10 g (m) of the ACMP (dried at 100 °C for 4 h) and filled a cylinder with water to the correct 200 mL mark. Then, we put the ACMP into the cylinder and determined the change in the water volume (V) as follows:

$$\text{Density of activated carbon} = \frac{m}{V} \quad (4)$$

2.5. Iodine Index of the Coal

We prepared 7 $\times$ 250-mL conical flasks. We added to each flask 0.05 g of the ACMP and 50 mL of an iodine solution with different concentrations and shook it carefully for 1 min, and then we filtered and titrated the remaining amounts of iodine with a 0.01 N $\text{Na}_2\text{S}_2\text{O}_3$ solution, with starch as an indicator. The experiments were conducted at room temperature (25 ± 1 °C).

The amount of iodine in the solution was calculated using the following equation [27]:

$$\text{iodine number (mg}\cdot\text{g)} = \frac{(V_1N_1 - V_2N_2)}{W} 126.93 f_g \quad (5)$$

where V_1 is the analyzed iodine volume (mL), V_2 is the volume of $\text{Na}_2\text{S}_2\text{O}_3$ used (mL), W is the activated carbon's weight (g), N_1 and N_2 are the iodine and $\text{Na}_2\text{S}_2\text{O}_3$ normality (N), respectively, f_p is the dilution factor, and 126.93 is the iodine amount corresponding to 1 mL of the $\text{Na}_2\text{S}_2\text{O}_3$ solution.

2.6. Determination of the Isoelectric Point of the ACMP

We prepared solutions of 0.1 M NaCl with the initial pH (pH_i) is increasingly adjusted from 0.94 to 8.35. We took 9 conical flasks of 100 mL capacity and added 0.05 g of the ACMP to each flask. Then, we consecutively added 100 mL of the solution with increasing pH (prepared as described above) to the conical flasks. We let the flasks stand for 48 h, then filtered the solutions and determined their pH (pH_f) again. We calculated the difference

between the initial pH (pH_i) and the equilibrium pH (pH_f) ($\Delta\text{pH} = \text{pH}_i - \text{pH}_f$), then plotted a graph showing the dependence of ΔpH on pH_i , and the intersection point of the curve with the coordinate at the value $\Delta\text{pH} = 0$ provided us with the isoelectric point to be determined.

3. Results and Discussion

3.1. Investigations on the Physical Properties, and Surface Characteristics of the MP and ACMP

SEM images of the MP and ACMP are displayed in Figure 2.

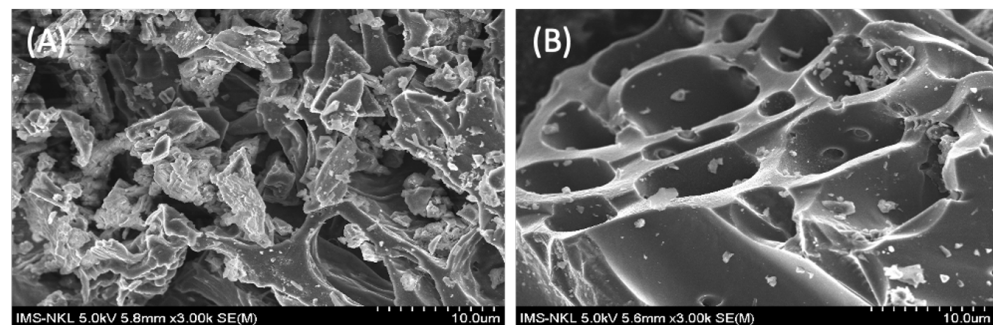


Figure 2. SEM image of the MP (A) and ACMP (B).

It can be observed in Figure 2 that mangosteen peel, after being activated by temperature and ZnCl_2 , showed remarkable surface morphology changes compared with the unactivated sample. Specifically, once being activated by ZnCl_2 (Figure 2), tubular capillaries were formed, creating many voids, increasing surface porosity, and enhancing adsorption efficiency.

The EDS analysis results of the MP and ACMP are summarized in Tables 2 and 3 and Figures 3 and 4. The results show that the ACMP possessed a carbon content mass of 81.45% and 86.54% by atom and an oxygen mass of 15.39% and 12.27% by atom. Meanwhile, the MP sample contained a carbon content mass of 68.81% and 76.86% by atom and an oxygen mass of 24.73% and 20.74% by atom. The increase in the carbon content (approximately 10%) and the decrease in the oxygen content showed that the MP material was successfully activated by the ZnCl_2 . This result is similar with the activated carbon derived from moringa leaves and activated by H_2SO_4 [36].

Table 2. EDX analysis of the MP.

Element	% Weight	% Atomic
C	68.81	76.86
O	24.73	20.74
Mg	0.70	0.39
P	0.38	0.16
Cl	0.13	0.05
K	4.76	1.63
Ca	0.50	0.17
Total	100.00	100.00

Table 3. EDX analysis of the ACMP.

Element	% Weight	% Atomic
C	81.57	86.74
O	15.31	12.22
P	0.20	0.08
Cl	0.22	0.08
K	2.70	0.88
Total	100.00	100.00

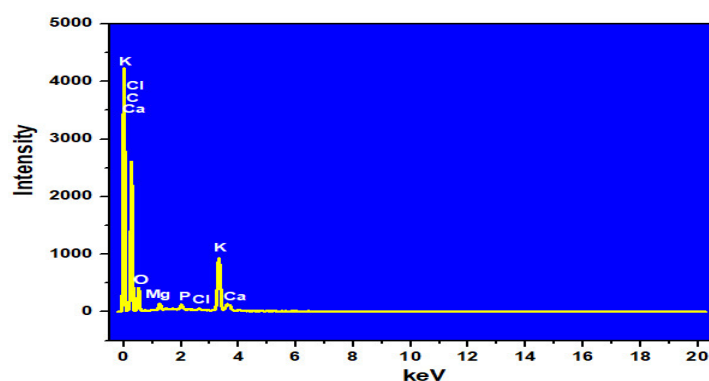


Figure 3. EDX spectrum of the MP.

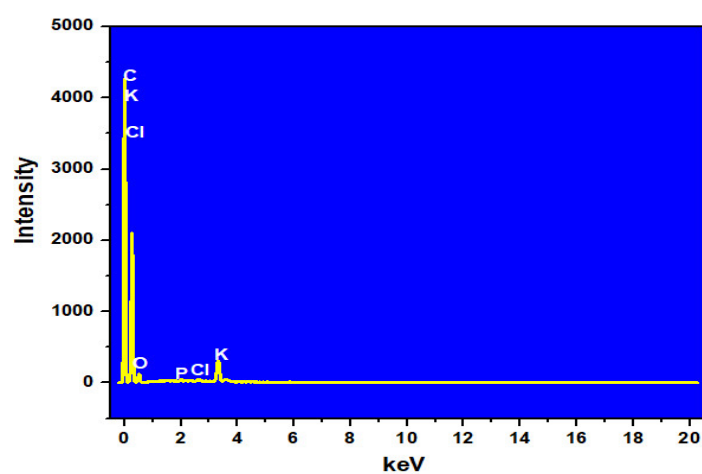


Figure 4. EDX spectrum of the ACMP.

The functional groups on the surface of the ACMP were examined through FT-IR infrared spectroscopy, as shown in Figure 5. Particularly, there was a broad-spectrum pattern at 3442.51 cm^{-1} attributed to the -OH group. Another spectrum pattern at 1627.56 cm^{-1} was ascribed to the valence vibration of the carbonyl group C=O (carboxylic). The infrared absorption of the symmetric CH group was shown at the peak spectrum positioned at 1388.03 cm^{-1} . The absorption spectrum located at 1043.74 cm^{-1} can be attributed to the oscillation of the C-O [36].

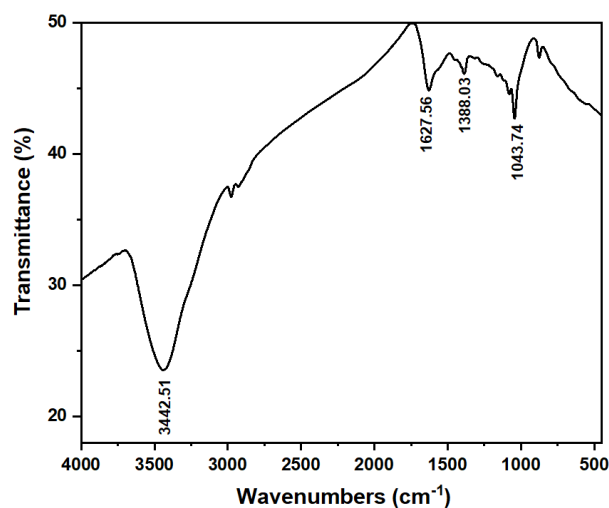


Figure 5. FT-IR analysis of the ACMP.

The surface properties and capillary structure of the ACMP were then studied by an N_2 adsorption–desorption method on a Micromeritics-3030 USA machine, as shown in Figure 6. It can be seen that the ACMP exhibited a type IV sorption isotherm curve with an H4 hysteresis loop. Type IV isotherms are common in porous adsorbents where capillary condensation occurs. Furthermore, H4 hysteresis loops are commonly found in adsorbents with particle sizes ranging from micropores to medium-sized pores. Figure 6 shows that the volume of the N_2 gas was adsorbed at the relative pressure P/P^0 (0–0.2), indicating the presence of micropore-sized material. The existence of material with a medium pore size is shown at the relative pressure P/P^0 in the range of 0.2–0.9. Furthermore, a small tail at the relative pressure P/P^0 , closer to 1.0 (>0.9), indicates the presence of a large pore-sized material. Therefore, the ACMP contains mainly medium-sized pores. This result is similar to the study published by Zhang et al. [35], where they authors made activated carbon from mangosteen peel and activated it with H_3PO_4 . On the other hand, from the BET measurement results, the average capillary volume was $0.279465 \text{ cm}^3/\text{g}$ and the average pore diameter was 2.66249 nm , which is consistent with the results of the nitrogen adsorption–desorption isotherms of the ACMP. The specific surface area of the ACMP was then measured as $419.8554 \text{ m}^2/\text{g}$.

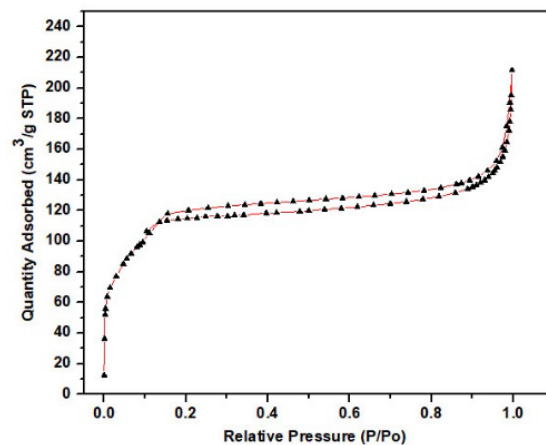


Figure 6. N_2 adsorption–desorption isotherms of the ACMP.

The XRD pattern of the ACMP is presented in Figure 7, showing two particular diffraction peaks in the range of $2\theta = 200\text{--}300$ and $2\theta = 400\text{--}500$. The presence of these peaks could be attributed to the lattice planes of 002 and 100 of the activated carbon. Therefore, the as-prepared ACMP possessed a graphite structure. This finding is consistent with the study on activated carbon's structure published by Zhang et al. [35].

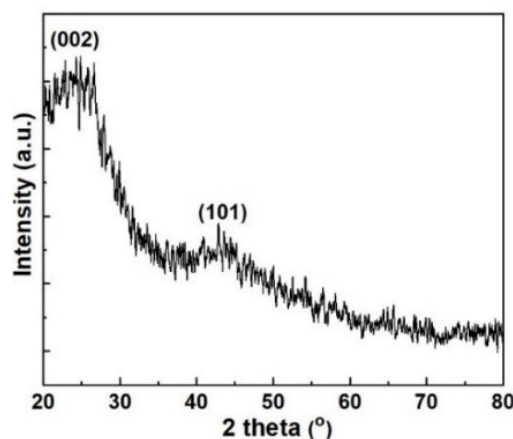


Figure 7. XRD pattern of the ACMP.

3.2. Isoelectric Point of the ACMP

The results of determining the isoelectric point of the ACMP are shown in Figure 8.

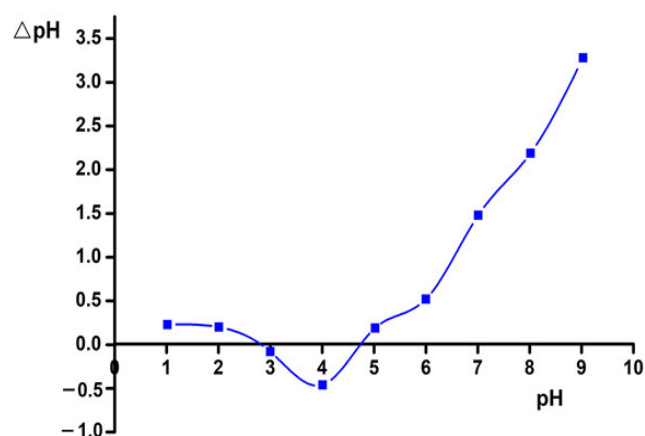


Figure 8. Graph of determining the isoelectric point of the ACMP.

From Figure 8, the point of zero charge (pH_{pzc}) of the ACMP was $\text{pH}_{\text{pzc}} = 5.34$. This verifies that the surface charge of the ACMP was positive when the pH was $< \text{pH}_{\text{pzc}}$, and it became negatively charged when the pH was $> \text{pH}_{\text{pzc}}$.

3.3. Physical Parameters and Iodine Index of the ACMP

The physical parameters of the ACMP are shown in Table 4.

Table 4. Results of the physical parameters and iodine index of the ACMP.

Ash content (%)	55.56 ± 1
Moisture content (%)	4.54 ± 0.03
Density (g/cm^3)	0.85 ± 0.05
Iodine index	825 ± 0.05

The ACMP has a low moisture content and density. In terms of mechanical properties, this would be favorable for the ACMP because a higher the moisture content is worse for the thermal engineering properties, reducing the strength and durability of the coal. The iodine index of the ACMP was 825 mg/g. The iodine index shows the adsorption capacity, as well as the porosity, of the fabricated activated carbon. This result was also within the range of the activated carbon that is suitable for practical applications in environmental treatments (500–1200 mg/g).

3.4. Investigations of the Influencing Effects on the CIP Adsorption Capability of the ACMP

3.4.1. Effect of pH

Determining a certain pH range for a certain process to achieve the highest efficiency is indispensable. As CIP is an antibiotic, it exists in a cationic form (CIP^+) due to the protonation of the amine group to below pH 5.9. When the pH is greater than 8.9, the CIP exists in an anionic form (CIP^-) because of the ionization of the carboxylic acid (Figure 9). In addition, the CIP can exist in a dipole form in the pH range of 5.9–8.9 (Figure 9). The dependence of the initial pH value on the CIP adsorption efficiency is shown in Figure 10A. Figure 10A also shows that the CIP adsorption efficiency of the ACMP materials gradually increased from the pH value of 2 to 6, and then gradually decreased from 6 to 10, with an especially rapid decline at the pH value of 7. This can be explained by the electrostatic interaction between the surface charge of the ACMP and the charge of the CIP. At a pH value of 6 (greater than the isoelectric point), the ACMP surface is negatively charged, and so there is strong cation adsorption (CIP^+) on the negative surface of the ACMP. The rapid

decline after pH 7 can be explained by the repulsive interaction between the negatively charged surface of the ACMP and the CIP^- . Gulen et al. [6] conducted a study on CIP adsorption in aqueous mediums using montmorillonite clay and found that this material exhibited its best CIP adsorption performance when the pH was equal to 7. Wang et al. [12] studied the adsorption of CIP in aqueous environments using activated carbon derived from bamboo, and they also noticed that the CIP adsorption efficiency could be prominently achieved at a pH value of 5.5. Thus, in this study, a pH of 6 was intentionally selected as the optimal adsorption condition for the CIP.

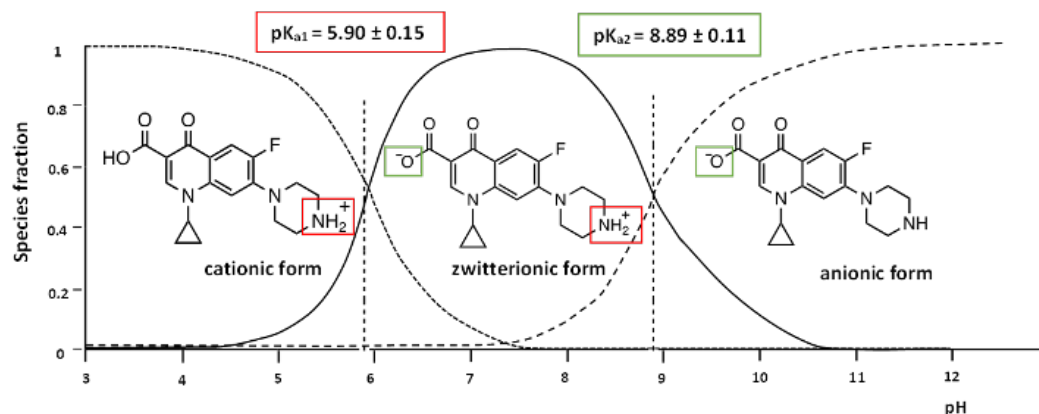


Figure 9. Existence forms of the CIP in solution [8].

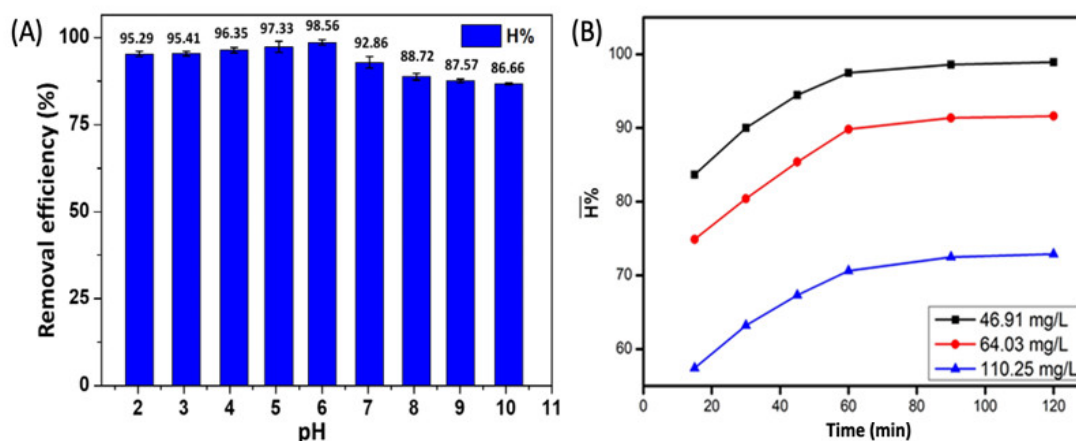


Figure 10. The effect of pH on the CIP adsorption of the ACMP (A) and of the reaction time on the CIP adsorption efficiency (B).

3.4.2. Effect of Reaction Time on Adsorption Equilibrium

As presented in Figure 10B, the graph lines show the dependence of the CIP adsorption efficiency on the reaction time at different CIP concentrations, and they both exhibited the same graph trend. Specifically, in the period from 15 to 120 min, the adsorption efficiency increased relatively quickly from 15 to 60 min, and it increased almost linearly. Then, it gradually stabilized in the period of 60 to 120 min because, initially, adsorption was fast due to the vacant adsorption sites on the adsorbent surface. After 120 min, adsorption slowed down due to the reduction in the number of available adsorption sites [9]. Therefore, adsorption equilibrium can be achieved in 60 min.

3.4.3. Effect of the ACMP Dosages on the CIP Adsorption Efficiency

It can be seen in Figure 11A that the CIP adsorption efficiency increases along with the increase in ACMP dosages, and this increase is linear in the range of the adsorbed ACMP dosages. This can be explained by the larger surface area and the increase in the number of adsorption sites. However, the adsorption capacity decreased because the adsorption

capacity was inversely proportional to the adsorbent dosage. However, as the range of the ACMP dosage increased from 0.075–0.125 g, the adsorption efficiency merely increased (from 94.95–97.22%). This is possibly due to the larger surface area and that more surface functional groups were available at higher concentrations of the adsorbent [37,38]. The decrease in adsorption capacity can be explained with the reduction in effective surface area of the adsorbent [39]. Therefore, we choose the ACMP dosage of 0.075 g (or the adsorption dosage/solution volume of 3 g/L) for further investigations.

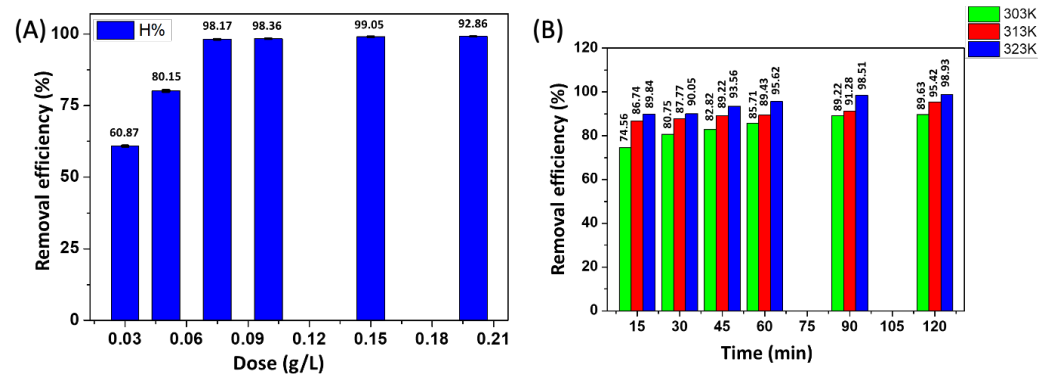


Figure 11. Effect of the adsorbent dosages (A) and temperatures (B) on the CIP adsorption efficiency.

3.4.4. Effect of Temperatures

From Figure 11B, it can be seen that when the temperature increased, the adsorption efficiency and the adsorption capacity both increased. This can be explained as follows: the adsorption of CIP on the ACMP was endothermic, and when the temperature was increased, the adsorption equilibrium shifted in a forward direction, which reduced the concentration of the adsorbent in the solution and led to an increase in the efficiency and adsorption capacity of the adsorption process. When studying CIP adsorption on adsorbent and onto activated bio-sorbents, Igwegbea et al. [8] found that the CIP adsorption process was thermodynamically favorable and was endothermic.

3.5. Isotherm Adsorption Model

3.5.1. Langmuir Isotherm Adsorption Model

To study isotherm adsorption models, experiments on the effect of the initial CIP concentration to the adsorption capacity of the ACMP were performed in the concentration range of 50 to 400 ppm (Table 1).

Langmuir adsorption, which was primarily designed to describe gas–solid phase adsorption, is also used to quantify and contrast the adsorptive capacity of various adsorbents. The Langmuir isotherm accounts for the surface coverage by balancing the relative rates of adsorption and desorption (dynamic equilibrium). Adsorption is proportional to the fraction of the surface of the adsorbent that is open, while desorption is proportional to the fraction of the adsorbent surface that is covered [39]. The Langmuir equation can be written in the following linear form:

$$\frac{C_0}{q_e} = \frac{1}{q_{\max}b} + \frac{C_e}{q_{\max}} \quad (6)$$

where q_e and $q_{\max}$ are the equilibrium adsorption capacity and the maximum adsorption capacity (mg/g), respectively, and b is the Langmuir constant. A Langmuir equation is characterized by the parameter R_L : $R_L = 1/(1 + b \cdot C_0)$. If $0 < R_L < 1$, then the adsorption is favorable, but if $R_L > 1$, then the adsorption is not favorable. If $R_L = 1$, then the adsorption is linear.

As seen in Figure 12A, the maximum adsorption capacity was determined as $q_{\max} = 29.76$ mg/g, with a constant $b = 0.173$ (L/g). The CIP adsorption of the ACMP is

well described according to the Langmuir isotherm adsorption model. This is expressed through the regression coefficient of the equation ($R^2 = 0.9986$).

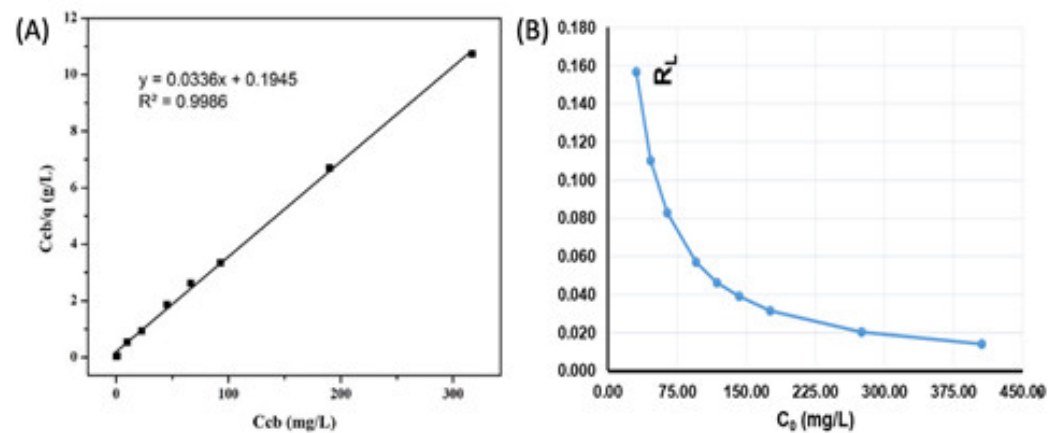


Figure 12. The dependence of C_{cb}/q on C_{cb} to the CIP adsorption parameter value R_L (A) versus C_0 (B).

The results of comparing the CIP adsorption capacity on different adsorbent materials (Table 5) show that the maximum adsorption capacity (q_{max}) of the ACMP for the CIP is relatively high. From the calculated parameter R_L value presented in the graph of Figure 12B, we can see that the values in the range of 0.154 to 0.014 are all less than 1, validating that the Langmuir isotherm adsorption model is appropriate with the CIP adsorption of the ACMP.

Table 5. The maximum CIP adsorption capacity (q_{max}) of the ACMP and other adsorbent materials.

No.	Adsorbents	Adsorption Capacity (mg/g)	Reference
1	Activated carbon from Lemna minor plant	25.00	[4]
2	Nano Fe_3O_4	24.00	[5]
3	Montmorillonite clay	128.00	[6]
4	Potato stems and leaves	23.36	[10]
5	Humic acid	13.64	[11]
6	Bamboo charcoal	36.02	[12]
7	Porous graphene–soy protein	160.00	[15]
8	ACMP	29.76	This work

3.5.2. Freundlich Isotherm Adsorption Model

The Freundlich isotherm is applicable to adsorption processes that occur on heterogeneous surfaces. This isotherm gives an expression which defines the surface heterogeneity and the exponential distribution of active sites and their energies.

The Freundlich isotherm adsorption equation has the form:

$$\ln q_e = \ln K_F + \frac{1}{n} \cdot \ln C_e \quad (7)$$

where K_F is the Freundlich adsorption constant and n is a constant that is always greater than 1.

Figure 13A describes the CIP adsorption process on the ACMP according to the Freundlich isotherm model. The Freundlich adsorption constant was measured as $K_F = 13.57$ (mg/g) (L/mg) $1/n$, with a constant value of $n = 6.73$. The correlation coefficient is $R^2 = 0.9421$. The large values of K_F and n indicate the good adsorption capacity of the material, and they show that there is an adsorption bond between the adsorbent and the adsorbate.

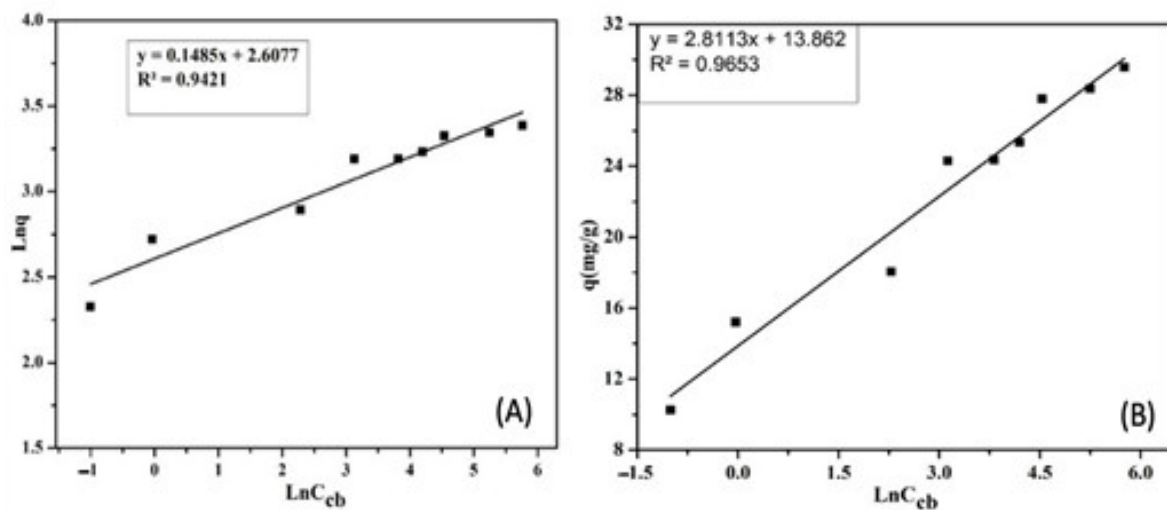


Figure 13. The dependence of $\text{Ln} \bar{q}$ on $\text{Ln} \bar{C}_{cb}$ to the CIP adsorption based on the Freundlich isotherm (A) adsorption model, and of $\text{Ln} \bar{q}$ on ϵ^2 based on the Tempkin isotherm (B) adsorption model.

3.5.3. Tempkin Isotherm Adsorption Model

The Temkin isotherm adsorption model is based on the assumption that the heat of adsorption is due to the adsorbent–adsorbent interaction (Temkin and Pyzhev, 1940), which has also been investigated. This isothermal adsorption model is applicable to chemisorption on a solid adsorbent and a liquid adsorbent. The Tempkin adsorption isotherm is represented by the following equation [39]:

$$q_e = B \ln K_T + B \ln C_e \quad (8)$$

where $B = RT/b_T$ and K_T is the Tempkin constant.

As seen in Figure 13B, the coefficient of determination R^2 was calculated as 0.9653. Therefore, the CIP adsorption process of the ACMP is propitious with the Tempkin isotherm adsorption model.

3.5.4. Elovich Isotherm Adsorption Model

The Elovich isotherm model is applied to the multilayer adsorption process. The Elovich adsorption isotherm model is expressed as [39]:

$$\ln \frac{q_e}{C_e} = \ln K_e q_m - q_e \quad (9)$$

where K_e is the Elovich constant and q_m is the Elovich maximum adsorption capacity.

The experimental results are presented in Figure 13A. As we can see in Figure 13A, there is a maximum adsorption capacity of $q_{\max} = 3.44 \text{ mg/g}$ and $K_e = 198.25$.

As seen in Figure 14B, the coefficient of determination R^2 of the Elovich isotherm adsorption model was determined to be 0.9508. Therefore, it is suitable for the CIP adsorption process of the ACMP.

3.5.5. Redlich–Peterson Isotherm Adsorption Model

The Redlich–Peterson isotherm model is a combination of the Langmuir and Freundlich isotherm models [38].

This isotherm model is an experimental isotherm model that integrates three parameters. It combines elements from both the Langmuir and Freundlich isotherm equations; therefore, the mechanism of adsorption is mixed and excludes monolayer adsorption [39].

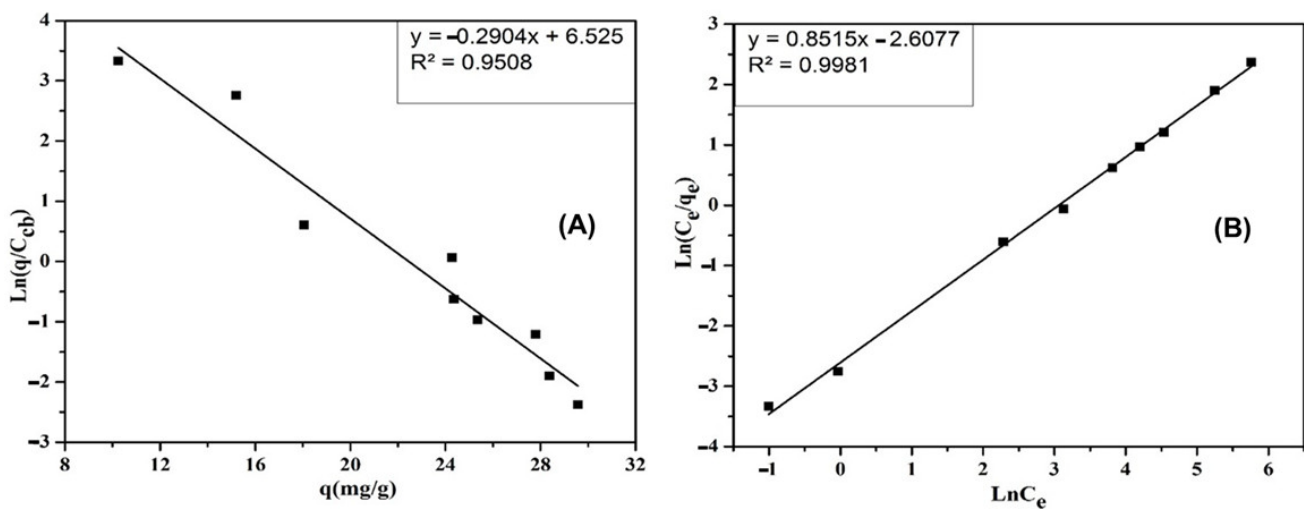


Figure 14. The dependence of $\ln(\bar{q}/\bar{C}_{eb})$ on $\bar{q}$, based on the Elovich isotherm (A), and of $\ln(C_e/q_e)$ on $\ln C_e$, based on the Redlich–Peterson isotherm (B) adsorption model.

The formula of this model is shown below:

$$q_e = \frac{AC_e}{1 + BC_e^\beta} \quad (10)$$

where A is the Redlich–Peterson isothermal constant (Lg^{-1}), B is a constant (Lmg^{-1}), β is the exponent between 0 and 1, C_e is the equilibrium concentration of the adsorbent (mg L^{-1}), and q_e is the equilibrium capacity of the adsorbent on the adsorbent (mg g^{-1}).

The linear form of the Redlich–Peterson isotherm is shown as:

$$\ln \frac{C_e}{q_e} = \beta \ln C_e - \ln A \quad (11)$$

We plotted a graph with $\ln(C_e/q_e)$ and with $\ln C_e$ to determine the β and A constants. The results are shown in Figure 13B. From the results in Figure 14B, the coefficient of determination R^2 of the Redlich–Peterson isotherm adsorption model is 0.9981. Therefore, it is suitable for the CIP adsorption process of the ACMP.

The values of constants calculated from the Langmuir, Freundlich, Tempkin, Elovich, and Redlich–Peterson isotherm adsorption models for the ACMP materials are shown in Table 6.

As seen in the results shown in Table 6, the coefficients of determination R^2 of the models—Freundlich, Elovich, Tempkin, and Redlich–Peterson—are 0.9986, 0.9421, 0.9653, 0.9508, and 0.9981, respectively.

The CIP adsorption on the ACMP follows the Langmuir > Redlich–Peterson > Elovich > Tempkin > Freundlich isotherm adsorption models. Therefore, it can be concluded that the CIP adsorption process of the ACMP is a monolayer and multilayer adsorption, on the conditions that the surface of the material is not uniform and that there is an interaction between the adsorbate and the adsorbent.

Table 6. Values of the constants calculated from the Langmuir, Freundlich, Tempkin, Elovich, and Redlich–Peterson isotherm adsorption models for the ACMP materials.

Isotherm Models		Parameters
Langmuir	b (L/mg)	0.173
	q _{max} (mg/g)	29.76
	R ²	0.9986
Freundlich	K _F (mg/g)·(mg/L) ^{1/n}	13.57
	N	6.73
	R ²	0.9421
Tempkin	K _T	138.49
	b _T (kJ·mol)	0.88
	R ²	0.9653
Elovich	q _{max} (mg/g)	3.44
	K _e	198.25
	R ²	0.9508
Redlich–Peterson	β	0.8515
	A (mg·L ^{−1})	13.57
	R ²	0.9982

3.6. CIP Adsorption Kinetics of the ACMP

Lagergren’s apparent first-order adsorption kinetics equation has the form:

$$\lg(q_e - q_t) = \lg q_e - \frac{k_1}{2.303} t \quad (12)$$

Lagergren’s apparent second-order adsorption kinetics equation has the form:

$$\frac{t}{q_t} = \frac{1}{k_2 q_e^2} + \frac{1}{q_e} t \quad (13)$$

where q_e and q_t are the adsorption capacities at the time of reaching equilibrium and at time t (mg/g), respectively; and k_1 and k_2 are the apparent first-order (time^{−1}) and second-order (g·mg^{−1}·time^{−1}) adsorption rate constants.

From the values of the parameters of the first-order kinematics equation in Table 7, the coefficient of determination R^2 ranges from 0.9812 to 0.9905. However, the equilibrium adsorption capacity values calculated from the kinetics equations are different from the experimental values. Therefore, the first-order kinetics equation is not suitable for the CIP adsorption by the ACMP (Figure 15).

Table 7. Values of the parameters from the first-order adsorption kinetics equation.

CIP Concentration (mg/L)	q _{t,exp} Experiment (mg/g)	q _{t,cal} Calculation (mg/g)	Constant k ₁ (min ^{−1})	R ²
46.91	15.24	4.86	0.0509	0.9875
64.03	19.17	6.13	0.0403	0.9812
110.25	25.96	10.10	0.0463	0.9905

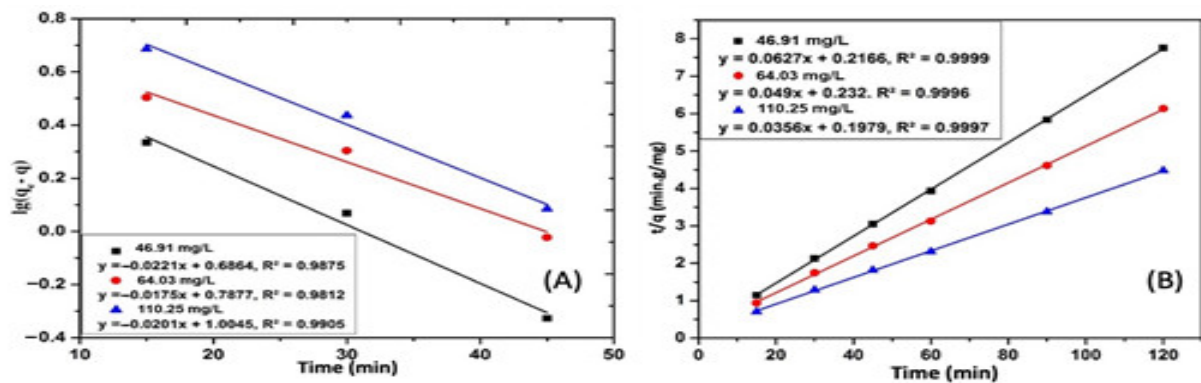


Figure 15. Graph of the first-order kinetics (A) and second-order kinetics (B) for the CIP.

In addition to the R^2 constant, the accuracy of the quadratic kinematics equation is also determined through the standard deviation $\% \Delta q$. The accuracy of the equation is determined by the standard deviation $\% \Delta q$ as the following equation:

$$\% \Delta q = 100 \times \sqrt{\frac{\sum [(q_{t,\text{exp}} - q_{t,\text{cal}}) / q_{t,\text{exp}}]^2}{n - 1}}$$

where n is the number of data points, $q_{t,\text{exp}}$ is the experimental adsorption capacity at time t , and $q_{t,\text{cal}}$ is the adsorption capacity at time t according to the kinetic equation.

From the values in Table 8, the coefficients of determination (R^2) are all 0.99. Further, the equilibrium adsorption capacity values calculated from the kinetics equations are very close to the experimental values, and the k_2 values are also approximately the same. The calculation results of the $\% \Delta q$ value in the table are low. Therefore, it can be concluded that the CIP adsorption process of the ACMP follows Lagergren's apparent quadratic kinetics equation. Therefore, the uptake of the CIP by the ACMP depends on the concentration of the CIP in the aqueous medium and the number of active sites on the adsorbent surface.

Table 8. Values of the parameters from the second-order adsorption kinetics equation.

CIP Concentration (mg/L)	$q_{t,\text{exp}}$ Experiment (mg/g)	$q_{t,\text{cal}}$ Calculation (mg/g)	Constant k_2 ($\text{min}^{-1} \text{L} \cdot \text{mg}^{-1}$)	R^2	$\% \Delta q$
46.91	15.24	15.95	0.0122	0.9999	1.04
64.03	19.17	20.41	0.0103	0.9996	3.02
110.25	25.96	28.09	0.0094	0.9997	3.73

This result is similar to that of Chinenye Adaobi Igwegbea [8], who studied the adsorption of ciprofloxacin from water and found that most of the CIP adsorption on materials obeyed the apparent quadratic kinetic equation of Lagergren, and only a few studies were first-order kinetics or obeyed the Elovich kinetic and Avrami fractional equations.

3.7. Activation Energy

To determine the activation energy (E) according to adsorption kinetics, we must firstly determine the value of h using the formula:

$$h = \frac{1}{b} \quad (14)$$

where b is the coefficient of the second-order kinetics equation and k is the rate constant of the second-order reaction. Then, the activation energy can be calculated using the following equation:

$$E = RT(\ln h - \ln k) \quad (15)$$

where R is the gas constant ($R = 8.314 \times 10^{-3} \text{ kJ/mol}\cdot\text{K}$) and T is the reaction temperature in Kelvin (K).

The activation energy calculated from the CIP adsorption process of the ACMP is displayed in Table 9. It can be seen that $E < 25 \text{ kJ/mol}$, and so the CIP adsorption of the ACMP is dominated by external diffusion. This result is similar to the result when conducting the adsorption of methylene blue using activated carbon made from moringa leaves [36].

Table 9. Activation energy parameters.

Concentration CIP (mg/L)	Second-Order Kinetics Equation	Constant k_2 (g/mg min)	h	E (kJ/mol)
46.91	$t/q_t = 0.0627 \cdot t + 0.2166$	0.0182	4.62	13.72
64.03	$t/q_t = 0.049 \cdot t + 0.232$	0.0103	4.31	14.96
110.25	$t/q_t = 0.0356 \cdot t + 0.1979$	0.0064	5.05	16.53

3.8. CIP Adsorption Thermodynamics of the ACMP

The isometric potential variation (ΔG^0), enthalpy (ΔH^0), and entropy (ΔS^0) of the adsorption process were calculated using the following equations:

$$K_D = \frac{q_e}{C_{cb}} \quad (16)$$

$$\Delta G^0 = -RT \ln K_D \quad (17)$$

$$\ln K_D = -\frac{\Delta G^0}{RT} = -\frac{\Delta H^0}{RT} + \frac{\Delta S^0}{R} \quad (18)$$

where K_D is the equilibrium constant.

The results of thermodynamic calculation on the CIP adsorption process of the ACMP are presented in Table 10.

Table 10. Thermodynamic parameters for the CIP adsorption.

T(K)	ΔG^0 (kJ/mol)	ΔH^0 (kJ/mol)	ΔS^0 (kJ/mol·K)
303	−0.845	66.957	0.223
313	−2.696		
323	−5.329		

It can be seen in Table 10 that the free energy variation value (ΔG^0) was determined as −0.845 to −5.329 kJ/mol and the entropy variation (ΔS^0) was 0.223 kJ/mol, proving that the CIP adsorption of the ACMP was a spontaneous process. The enthalpy variation (ΔH^0) was measured as 66,957 kJ/mol, verifying that the adsorption process was endothermic.

During adsorption process, $\Delta H^0 < 25 \text{ kJ}\cdot\text{mol}^{-1}$, and the van der Waals forces are majorly responsible for physical adsorption. If $\Delta H^0 = 40\text{--}200 \text{ kJ}\cdot\text{mol}^{-1}$, the chemical bonding mainly leads to chemisorption [40]. The results of studying the CIP adsorption process of the ACMP follow the apparent second-order kinetics equation and the adsorption isotherm models of Tempkin, Elovich, and Redlich–Peterson, indicating that the adsorption of the CIP on the ACMP can be dominated by chemisorption, which is in agreement with the calculation results of $\Delta H^0 = 66.957$ of the CIP adsorption on the ACMP. The results of the thermodynamic calculation from the CIP adsorption process by the ACMP are similar to those reported by Zhe Zhang et al. [35], wherein the adsorption process of methylene blue in

the environment on activated carbon materials made from mangosteen peel also exhibited an endothermic, spontaneous process which is governed by chemisorption. However, this is a different result from the results achieved by the authors when studying the CIP adsorption on biological activated carbon materials where they found that the adsorption process was a physical adsorption [7].

4. Conclusions

In this study, activated carbon derived from mangosteen peel (abbreviated as ACMP) was successfully fabricated via physical activation, followed by chemical activation with ZnCl_2 , with a mass ratio of MP: ZnCl_2 as 2:1, an activation temperature of 500 °C, and an activation time of 180 min. This as-prepared ACMP possessed a graphite and porous structure, with a specific surface area of 419.8554 m^2/g , an iodine index of 825 mg/g , and an isoelectric point equal to 5.34. A total of 98% CIP adsorption was achieved at the initial concentration of 50 ppm. In addition, the CIP adsorption process of the ACMP followed the Langmuir, Freundlich, Tempkin, Elovich, and Redlich–Peterson isotherm adsorption models, which determined that the maximum CIP adsorption capacity on the ACMP was $q_{\text{max}} = 29.78 \text{ mg}/\text{g}$. The CIP adsorption process of the ACMP followed the apparent quadratic kinetics and spontaneous, endothermic, physical, and chemical adsorption. The adsorption rate was governed by membrane diffusion. Therefore, the ACMP activated with ZnCl_2 could be a promising material for antibiotic treatment. It is an inexpensive and viable material for removing antibiotics from contaminated wastewater.

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